Heterogeneous Graph Learning for Article Recommender System

DS300.Q11 - Final Project

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DS300 Final Project Report

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Motivation: Why News Recommendation?

In the digital era, news platforms face **Information Overload**:

- **User Experience:** Navigating thousands of daily articles is overwhelming.
- **Retention:** Personalized content increases user engagement and platform stickiness.
- **Monetization:** Better engagement leads to higher ad revenue and subscriptions.

Target: Build a system that understands both *Content semantics* and *User behavior*.

Research Objectives

Our goal is to develop a recommendation engine for Vietnamese news:

- **Graph Learning:** Leverage GNNs to model complex user-article interactions.
- **Multimodal Fusion:** Incorporate Social (replies) signals.
- **Vietnamese NLP:** Optimize PhoBERT embeddings for news semantics.

Problem Statement

Challenge: How to recommend relevant news articles to users?

Key Challenges:

1. **Cold Start:** New users/articles have no interaction history
2. **Sparse Data:** Most users read very few articles (99.9% sparsity)
3. **Vietnamese Language:** Limited NLP resources for Vietnamese text
4. **Interaction Limitation:** Don't have access different kind of interactions.

Our Approach: Graph-based Contrastive Learning with PhoBERT embeddings

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VnExpress Data Crawling

Data Source: VnExpress.net (largest Vietnamese news portal)

Categories Crawled:

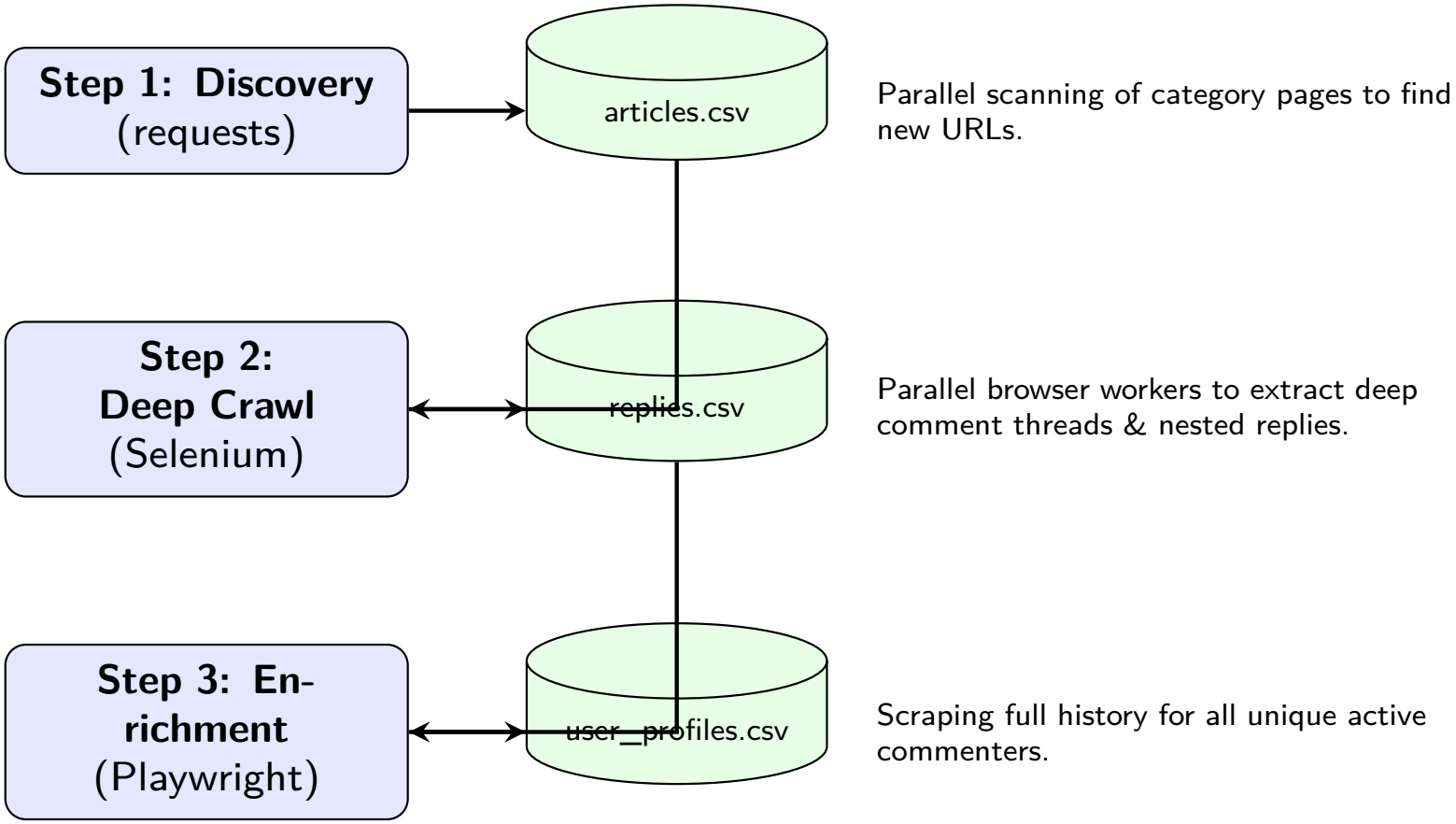
- 'Thế giới'
- 'Giáo dục'
- 'Thể thao'
- 'Thời sự'
- 'Kinh doanh'
- 'Khoa học công nghệ'

Collected Data:

- Articles: title, content, author, publish date
- Metadata: category, tags (per article)
- Comments: user ID, comment text, reactions, replies
- User profiles: user ID, username, join date

Data Collection Pipeline: 3-Step Workflow

Our crawler implements a resilient 3-stage architecture to handle high-volume news data:



Pipeline Orchestration & Efficiency

- **Parallelism:** `Step 2` (Selenium) is resource-intensive. We split the workload across N parallel browser instances using `multiprocessing`.
- **Resiliency:**
  - **Caching:** Avoids redundant HTTP requests for static content.
  - **Checkpointing:** Atomic writes to CSV allow resuming from the last article in case of failure.
  - **Headless Execution:** Minimizes resource overhead in production environments.
- **Tool Stack:**
  - **Requests:** High-speed metadata extraction.
  - **Selenium:** Handling complex JS (Load More buttons).
  - **Playwright:** Modern, fast engine for user history scraping.

Dataset Overview: High-Level Statistics

Raw Behavioral Data:

- **3,685** Articles
- **42,050** Comments
- **6** News Categories
- **19,842** User Profiles

Graph Interaction Set (G1-G3):

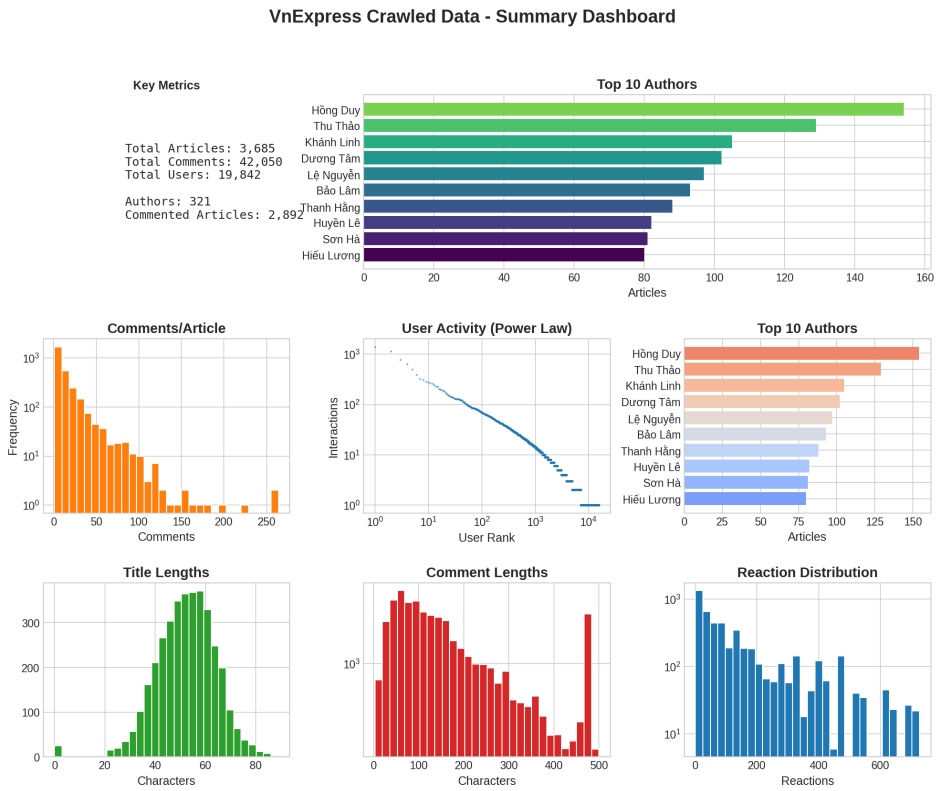
- **13,117** Unique Users
- **3,645** Active Articles
- **38,505** Total Interactions
- **Sparsity: 99.919%**

Dataset Overview (VnExpress)

Statistics:

- **Users:** 13,117
- **Articles:** 3,645
- **Interactions:** 38,505
- **Sparsity:** 99.92%

The dataset is highly sparse, typical of news domains where users read only a tiny fraction of daily content.



Data Sample: Interaction & Content

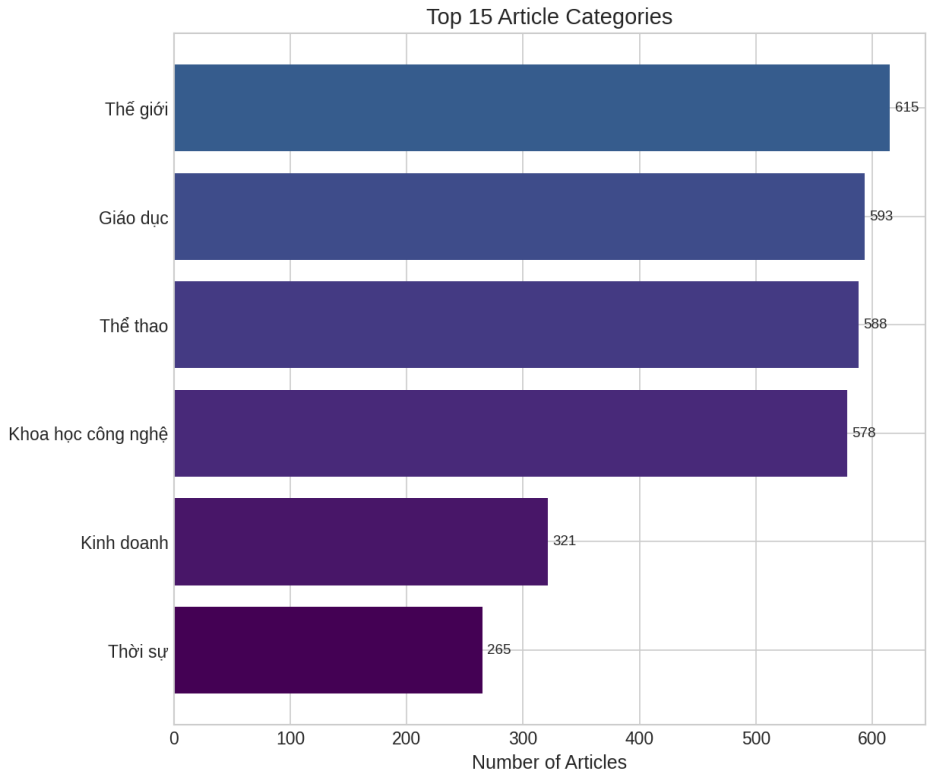
1. **Content Sample (Article)**
"Chinh phục loạt 'ông lớn' công nghệ sau hơn 500 lần bị từ chối"
→ **Tags:** [Meta, Amazon, Apple, Tesla, kĩ sư công nghệ]

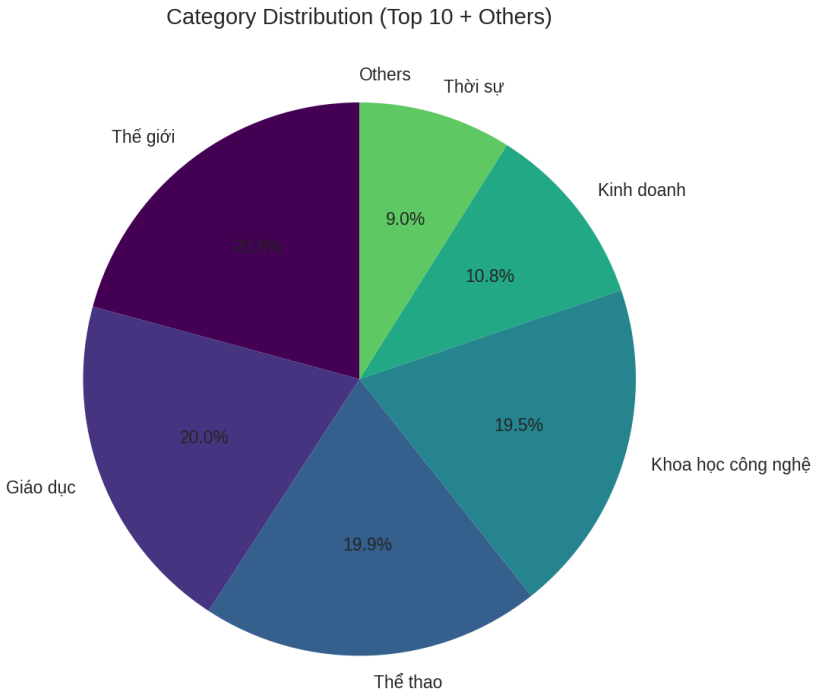
2. **Social Sample (Comment/Reply)**
   - **User A:** "Nếu giỏi thì cố gắng mua lại mấy công ty này..."
   - **User B:** (Reply to A) "Mua lại Meta chắc cần nghìn tỷ USD..."
   ⇒ Social Connection detected

Feature Mapping

- **Article** → PhoBERT Embedding ($d = 64$) → Node Feature.
- **User** → Interaction aggregation → Node Embedding.
- **Reply** → User-User Edge in the Heterogeneous Graph.

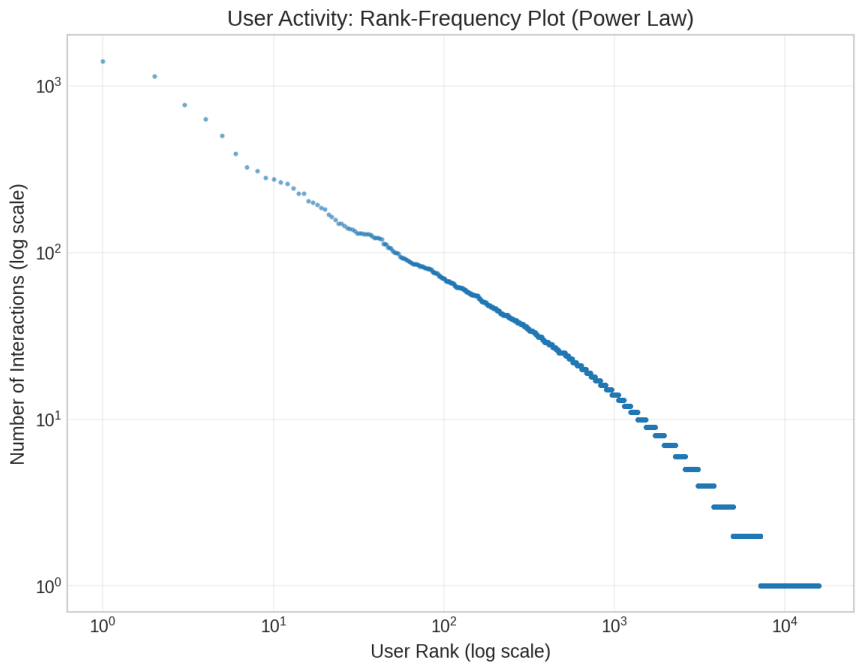
Category Distribution (VnExpress)

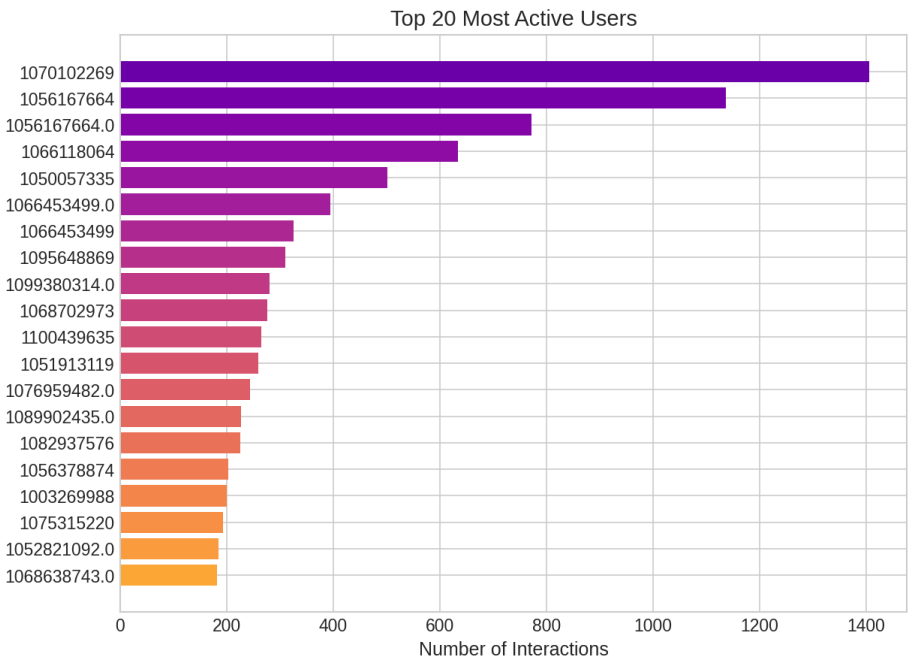




- **Thế giới** and **Thời sự** are the dominant categories.
- Skewed distribution reflects editorial priorities.

User Engagement Patterns (1/2)

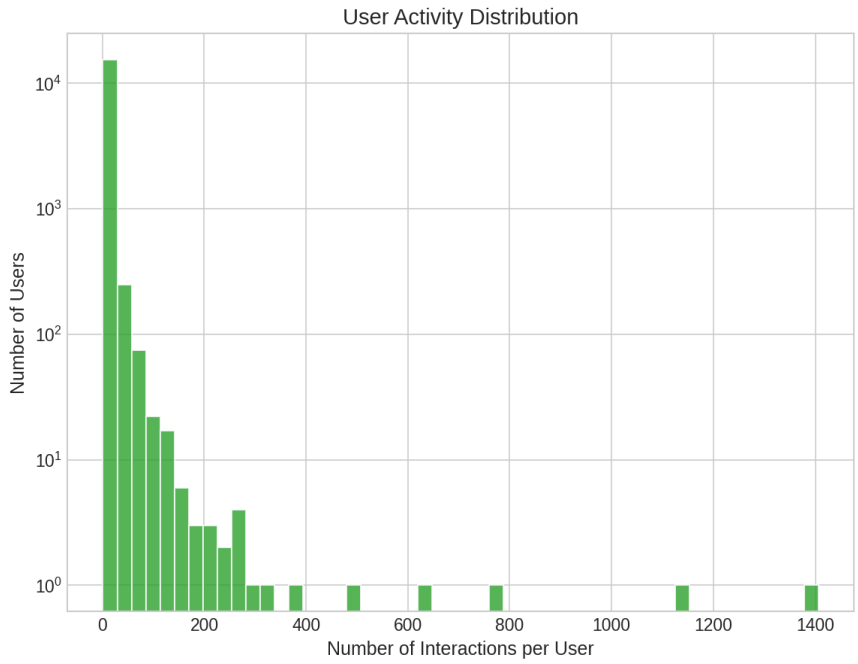


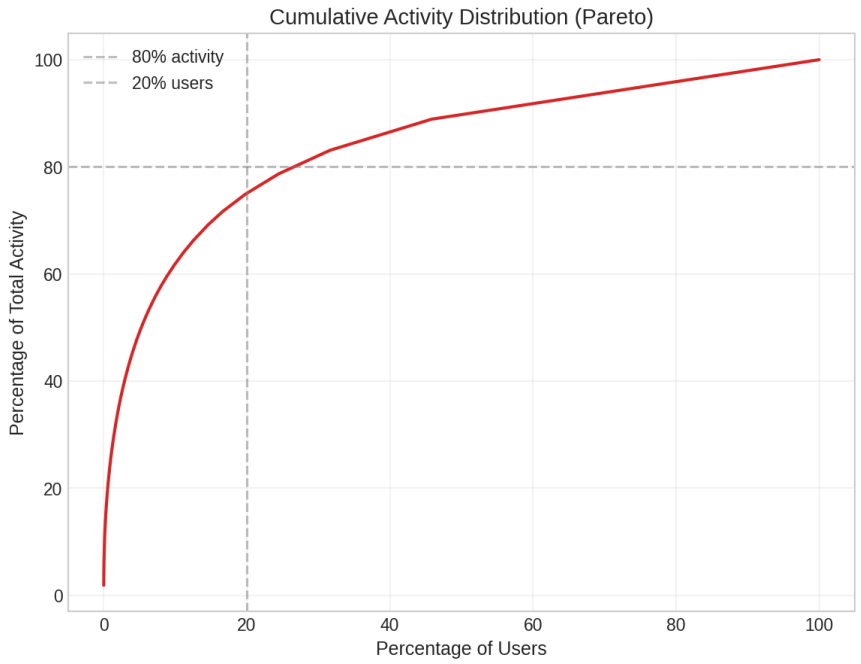


Distribution Analysis

Article popularity follows Zipf's law. Top 10 users account for 15% of all interactions.

User Engagement Patterns (2/2)

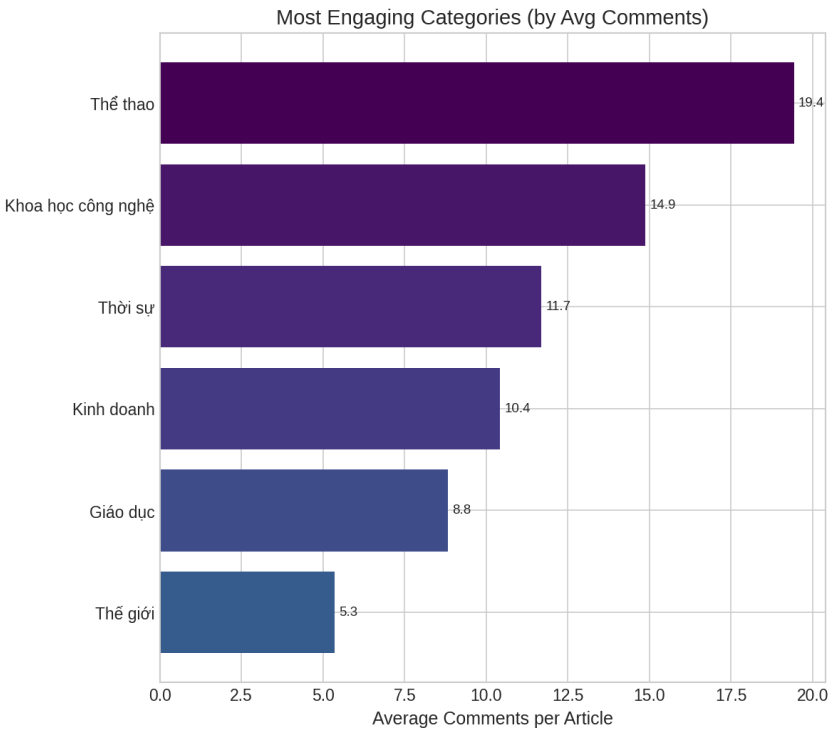


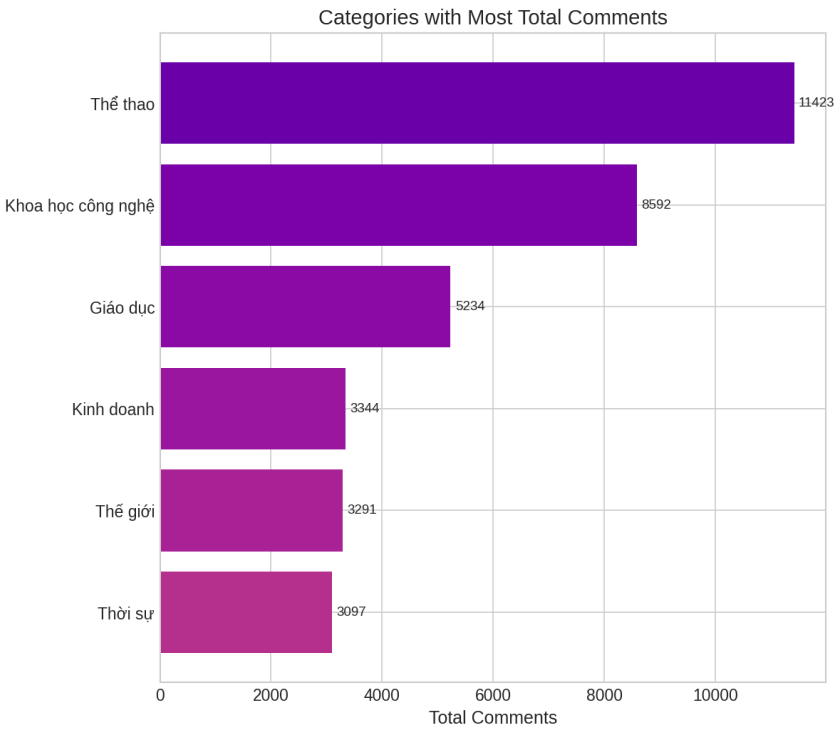


Power Law Observation

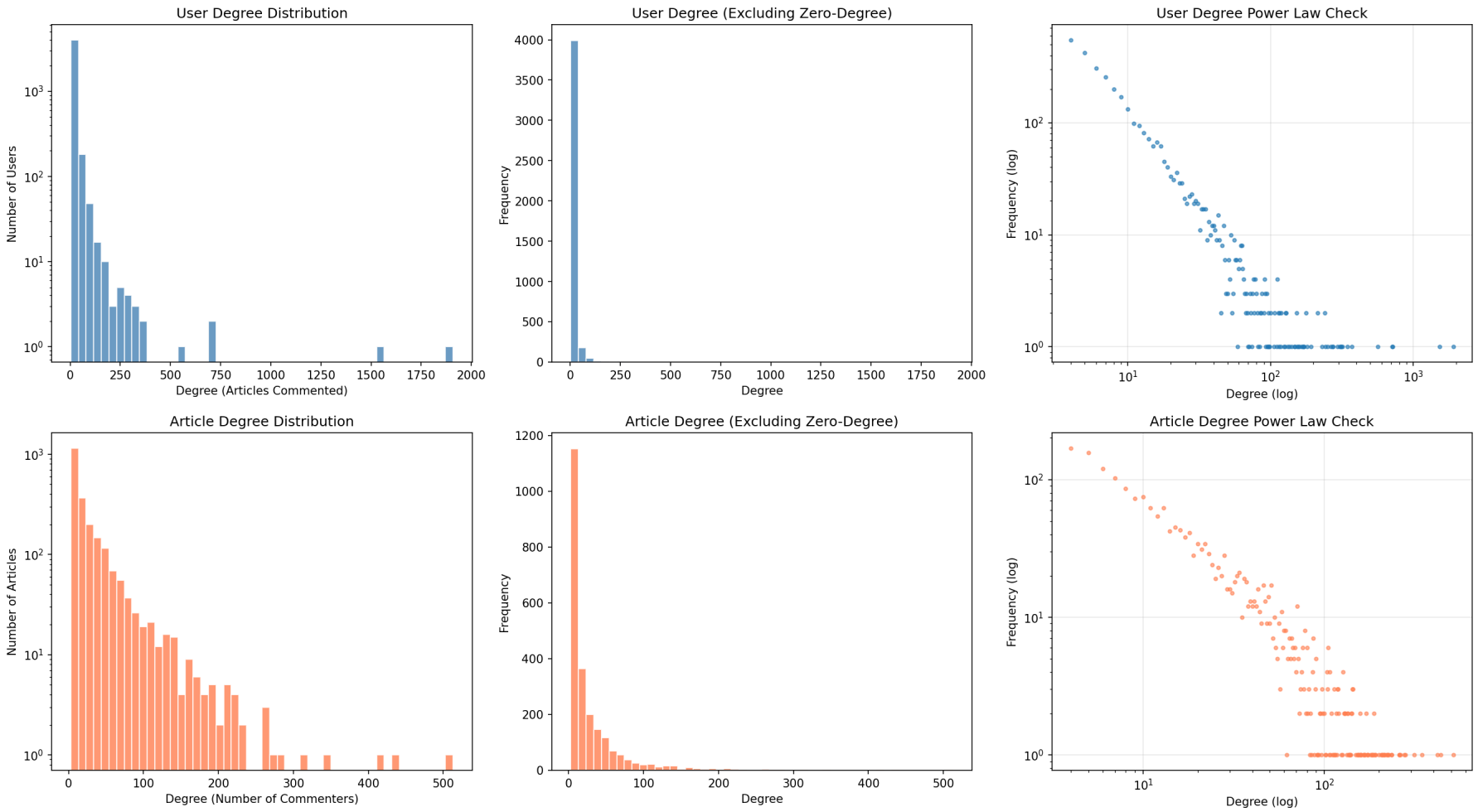
Long-tail distribution: 80% of users have fewer than 10 interactions.

Engagement by Category



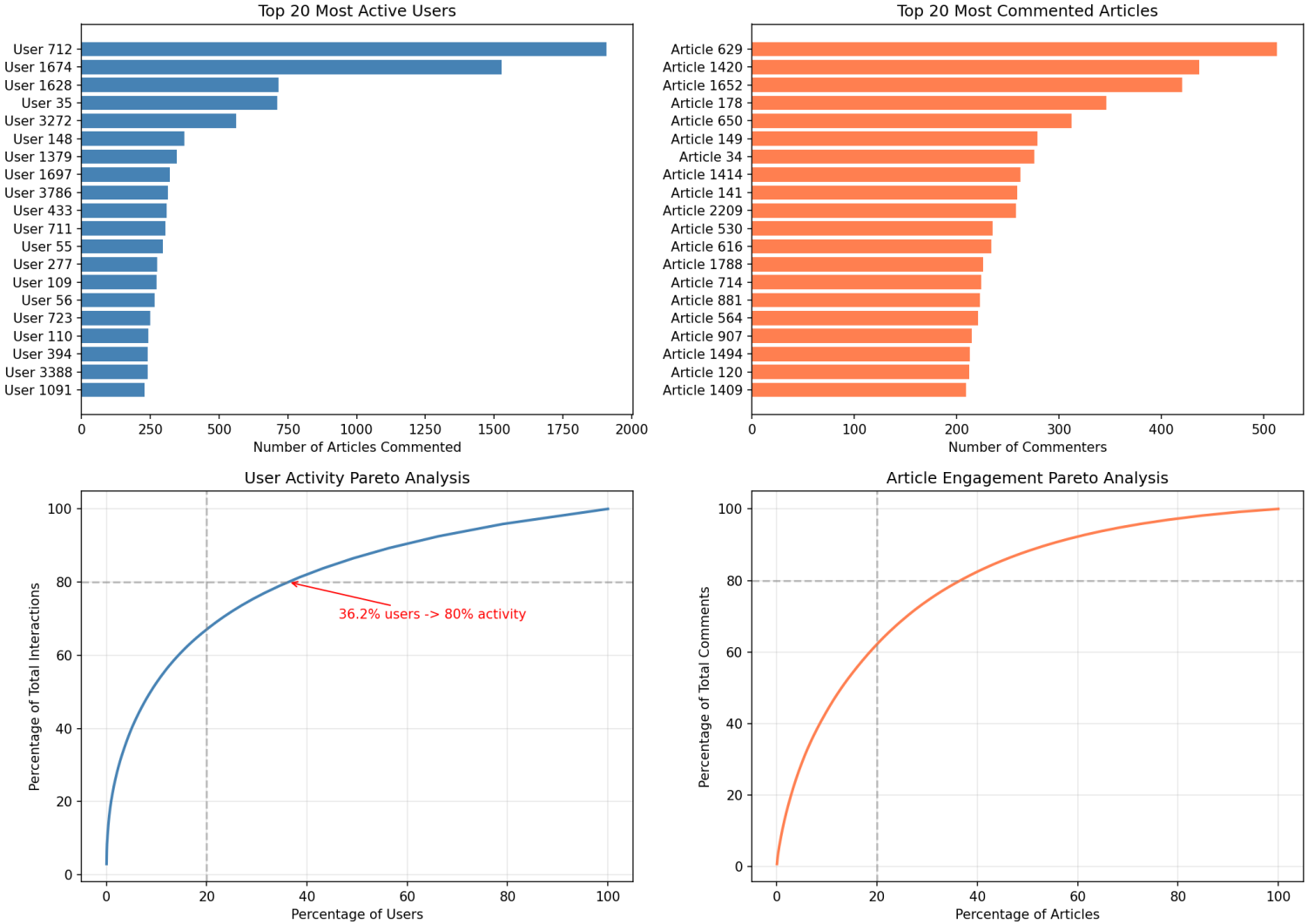


Graph Degree Distribution



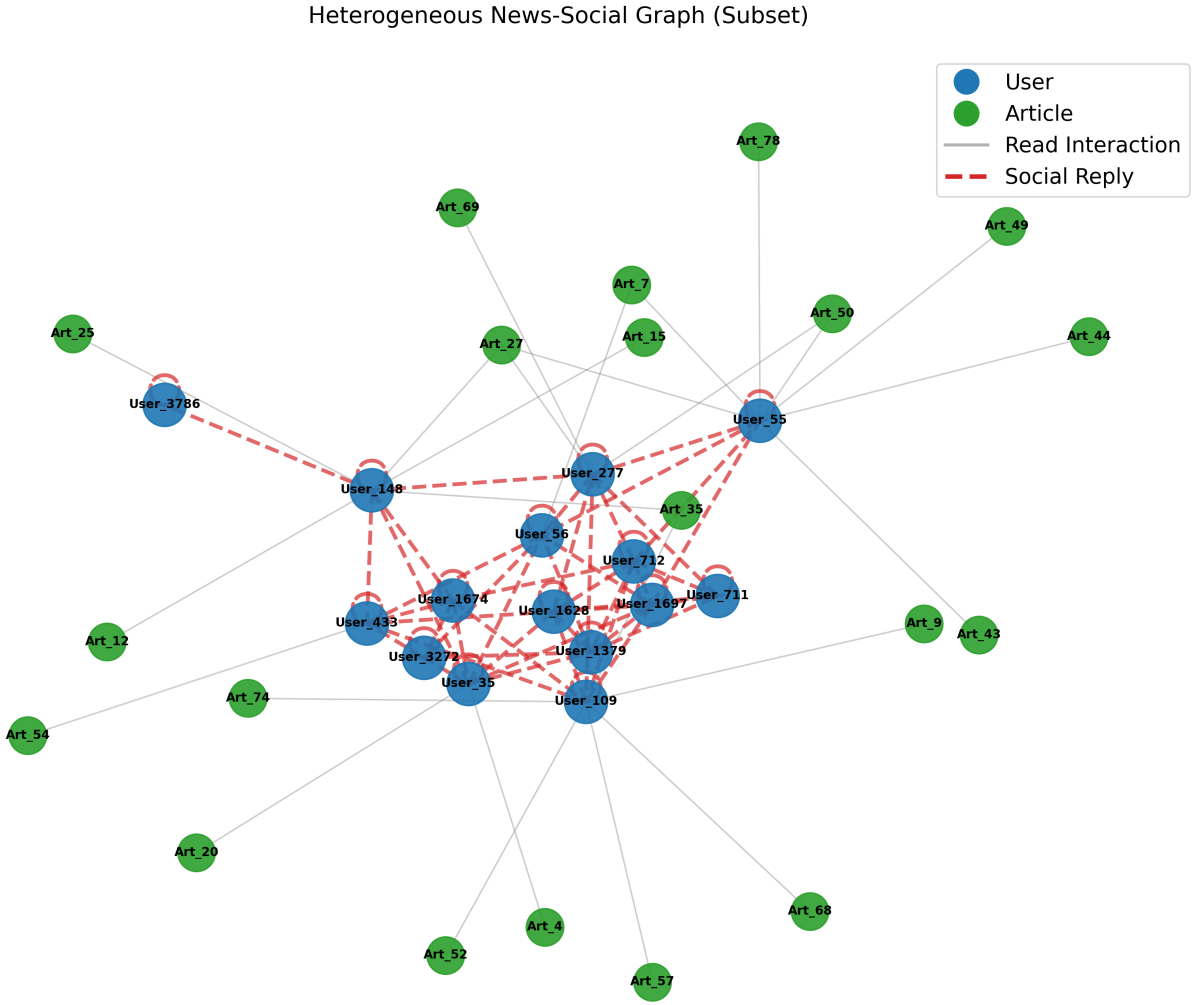
- User degrees peak at very low values (high cold-start risk).
- Article degrees are more distributed but still sparse.

Graph Connectivity & Density



Graph Topology

Visualizing the User-Article-Comment bipartite graph structure.



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The Sparsity Challenge

Problem: Interactions are extremely sproparse (99.919%), making it hard for models to learn.

Sparsity Calculation

$$\text{Sparsity} = 1 - \frac{38,505 \text{ (Interactions)}}{13,117 \text{ (Users)} \times 3,645 \text{ (Articles)}} = \mathbf{99.919\%}$$

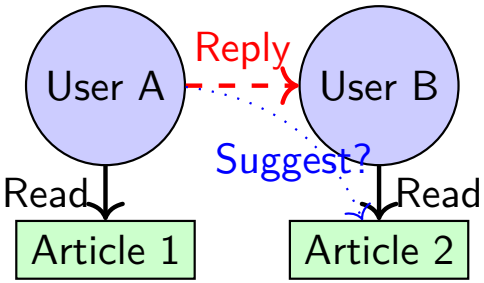
Solution: Expanding the Graph Structure

We progressively enrich the graph to alleviate sparsity. We define **3 Graph Tiers** for experimentation:

- **Graph G1 (Bipartite Baseline):**
  - Nodes: Users, Articles.
  - Edges: *Comment*.
  - *Standard CF setting.*
- **Graph G2 (Social Expansion - "The Increase"):**
  - **+ User-User Edges:** Created via "Reply" interactions.
  - *Hypothesis: Users who converse share interests.*
- **Graph G3 (Semantic Expansion):**
  - **+ Category Nodes:** Users connect to Categories they read.
  - *Hypothesis: High-level hubs reduce path length.*

Core Innovation: Social Fusion (Graph G2)

Idea: Use the "Reply Network" to bridge gaps between users.



The Logic:

- **Direct:** User A likes Article 1.
- **Social:** User A replies to User B $\rightarrow$ *Shared Interest*.
- **Inference:** User A is likely to enjoy Article 2 (read by B).

Impact on Sparsity

By adding User-User edges (G2), the GNN can propagate "interests" even if User A has only read 1 article.

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Current SOTA Approaches

We benchmark against three main categories of modern Recommender Systems:

1. **Graph Neural Networks (GNNs):**
   - **LightGCL (2023):** Efficient Global Context Learning via SVD for better structural alignment.
   - **XSimGCL (2023):** Extreme Simple Graph Contrastive Learning using noise perturbation instead of graph augmentation.
   - **LightGCN (2020):** Simplified GCN by removing non-linearities. Current industry standard for efficiency.
2. **Contrastive Learning (CL):**
   - **SimGCL (2022):** Augments graph structure with noise to learn robust representations. Handles sparsity better than pure GNNs.
3. **Content-Based & Hybrid:**
   - **PhoBERT:** State-of-the-art BERT model for Vietnamese, capturing deep semantic text features.

State-of-the-Art Baselines: Contrastive Learning

1. LightGCL (Global Context)

Problem: Standard GNNs only capture local neighbors.
Solution: Learn **Global Context** via SVD ($G = U_k \Sigma_k V_k^T$).
Why News? Breaks "Info Cocoons" by recommending distinct but related topics.

2. XSimGCL (Extreme Simple)

Problem: Graph augmentation (drop edge) is slow.
Solution: Add random **Noise** (Δ) to each embeddings layers: $e' = e + \Delta$.
Benefit: Fast (34s/epoch) & Robust regularization.

Both methods outperform traditional GCNs by improving representations.

HGRec: Heterogeneous Graph Neural Network

Core Idea: Real-world graphs have multiple node/edge types (e.g., User-Reply-User vs. User-Comment-Article). Standard GNNs ignore this distinction [1].

1. Meta-path Aggregation

Separately aggregates neighbors for each relation type:

$$h_u^{\Phi} = \text{AGG}\{h_v | v \in \mathcal{N}_u^{\Phi}\}$$

Example: Captures "Social" signals separately from "Interest" signals.

2. Semantic Fusion

Combines type-specific embeddings using Attention:

$$h_u^{Final} = \sum_{\Phi} \beta_{\Phi} \cdot h_u^{\Phi}$$

Benefit: Learns which relation is more important (e.g., Social > Interest for inactive users).

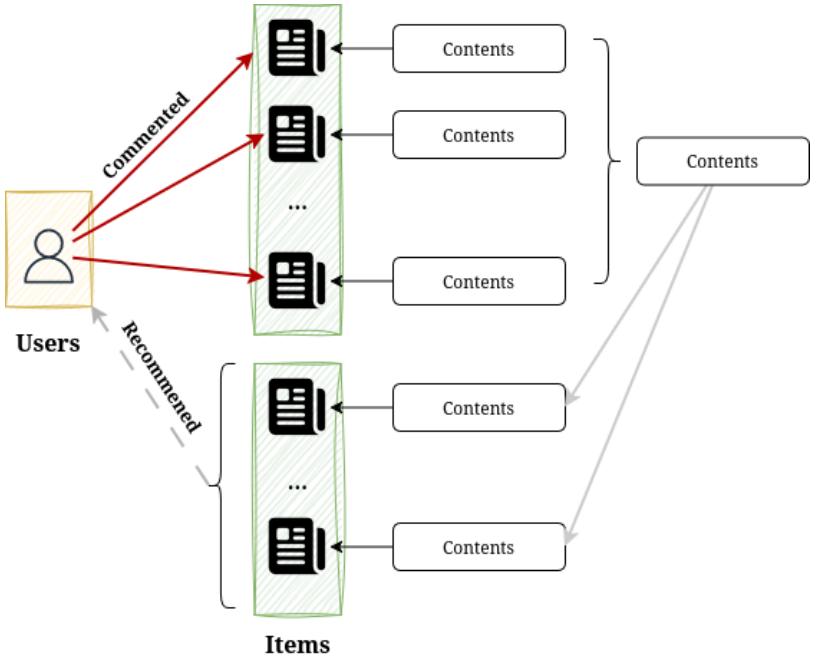
Significance: This mechanism forms the backbone of our **HGN** and **HCL** models.

Approach 1: Content-Based Filtering (CB)

Concept: Recommend items similar to what the user liked in the past.

Process Flow

- **Input:** Title, Abstract.
- **Method:** PhoBERT $\rightarrow$ Dense Vectors.
- **Metric:** Cosine Similarity.
- **Output:** Top-K semantically similar items.

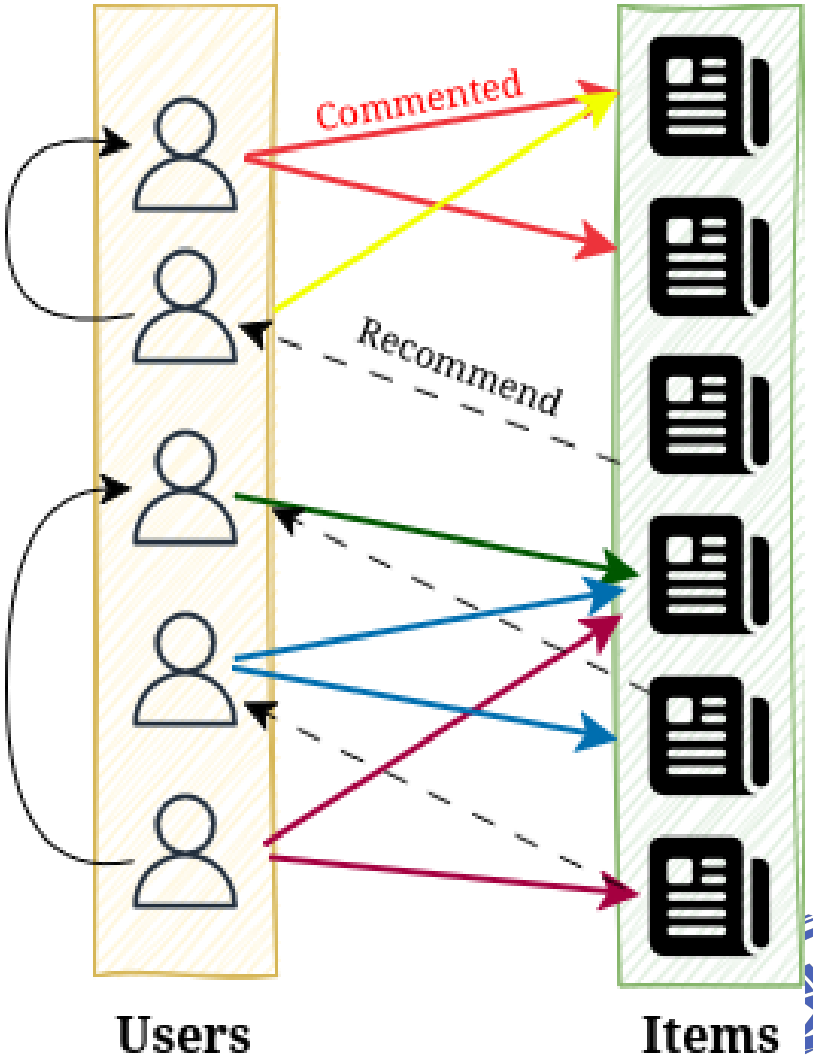


Approach 2.1: User-User Collaborative Filtering

Concept: Leverage "Crowd Wisdom" - "Users like you also liked..."

Mechanism

1. Identify similar users based on interaction history (e.g., Cosine Similarity).
2. Aggregate items consumed by these neighbors.
3. Recommend items unseen by the target user.

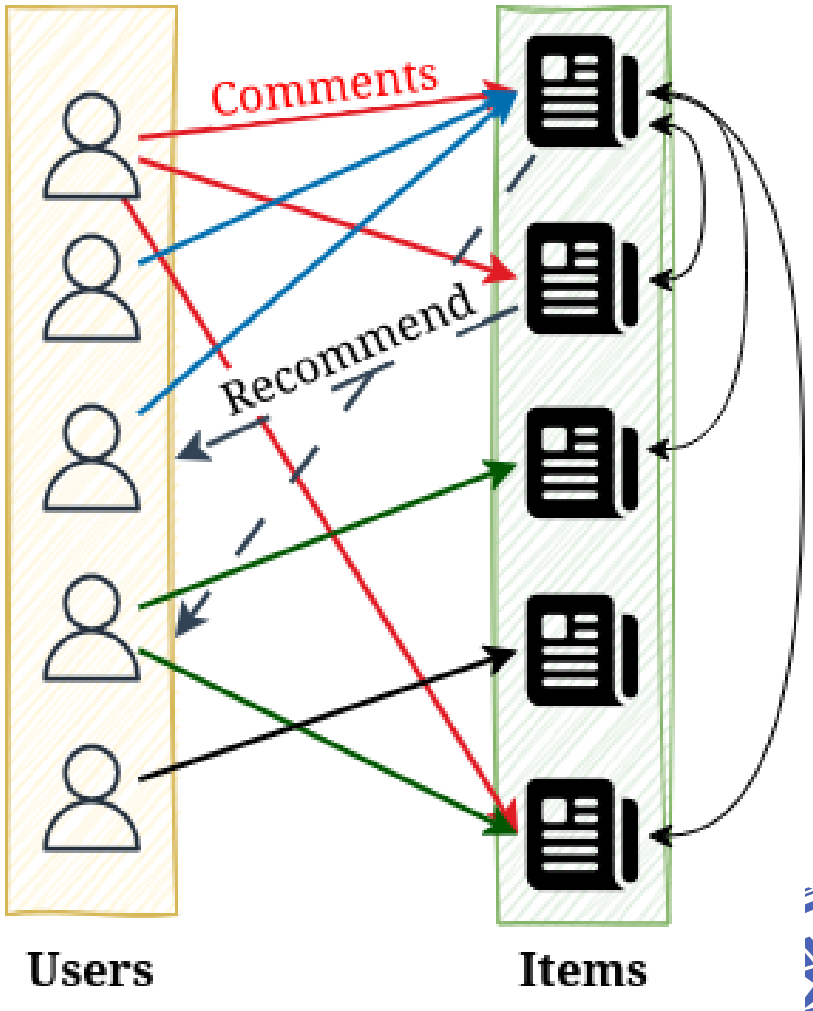


Approach 2.2: Item-Item Collaborative Filtering

Concept: "Users who bought X also bought Y"

Mechanism

1. Calculate similarity between items based on co-occurrence in user histories.
2. If a user likes Item A, recommend Item B (where A and B are similar).
3. More stable than User-User as the item set is strictly static.



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Inspiration & Theoretical Foundation

Where did this idea come from? Our architecture is an adaptation of the **HGRec** framework [1], optimized with LightGCN **he2020lightgcn** mechanics.

1. HGRec (Heterogeneous Graph Neural Network)

Heterogeneous GNN for Recommendation

- **Key Idea:** Users and items have multiple types of relationships (e.g., Comment vs Reply).
- **Mechanism:** Use relation-specific aggregation steps.

2. LightGCN

Simplified GCN for Recommendation

- **Key Idea:** Non-linearities (ReLU) are effectively redundant for collaborative filtering.
- **Mechanism:** Linear propagation + Layer averaging.

Motivation: Why a Heterogeneous GNN?

The Problem with Standard GNNs (LightGCN):

- They treat the graph as **Homogeneous**: All nodes are just "IDs", all edges are just "links".
- **Failure Case:** A "Reply" (social trust) is treated identically to a "Comment" (explicit interest).

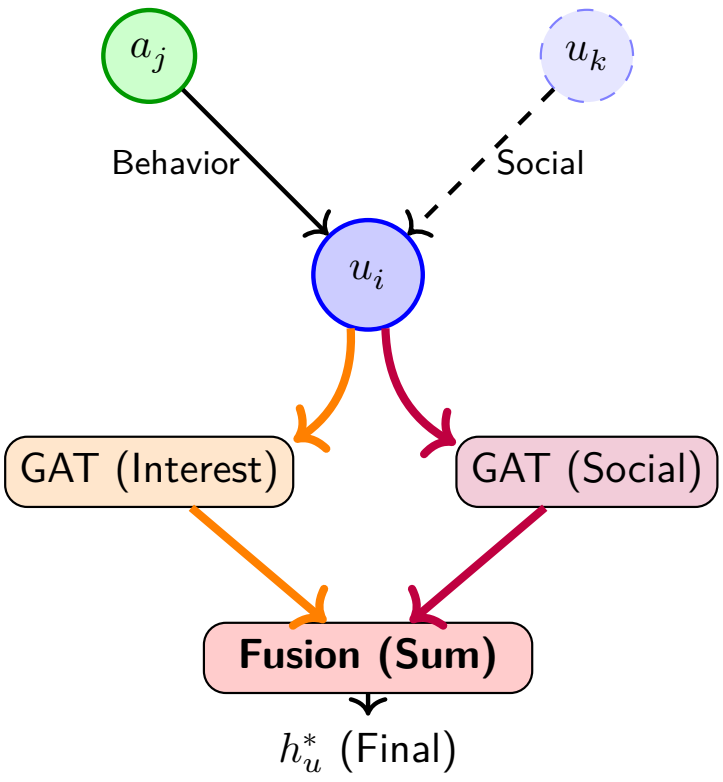
The VnExpress Reality:

- **Multiple Node Types:** Users $\neq$ Items.
- **Multiple Edge Types:**
  - `User-Comment-Item`: Explicit Interest.
  - `User-Reply-User`: Social Influence / Trust.

The HGN Solution

We need an architecture that aggregates **Social Signals** separately from **Content Signals** before fusing them.

HGN: Multi-Aspect Heterogeneous GNN



Core Innovation: Dual-path Propagation

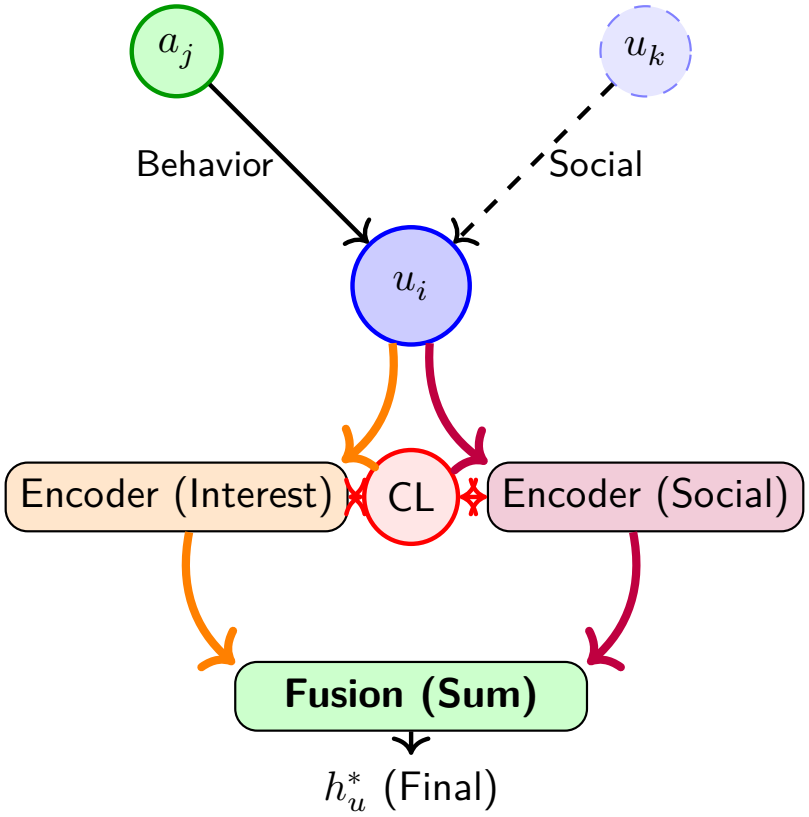
- **Path 1 (Interest):** history ($\mathcal{G}_{u-a}$) via **GAT**.
- **Path 2 (Social):** friends ($\mathcal{G}_{u-u}$) via **GAT**.
- **Fusion:** Combines behavioral and social signals by simple summation:

$$h_u = h_{int} + h_{soc}$$

Alternative Component: HCL (Failed Experiment)

Idea: Combine Heterogeneity (Social + Interest) with Contrastive Learning + LightGCN.

Architecture Diagram:



Note: Contrastive Loss proved unstable in this sparse setting.

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Experimental Setup

Models ($N = 6$):

- **Baselines:** LightGCN, HGN
- **Contrastive:** LightGCL, SimGCL, XSimGCL (SOTA), HCL (Ours)

Training config:

- **Epochs:** 100
- **Batch Size:** 2048
- **Loss:** BPR + InfoNCE (for CL models)

Data Protocols:

- **Full (Cold-Start):** 80/20 random user split.
- **LOO (Mainstream):** Temporal Leave-One-Out.

Graph Tiers:

- **Strict G1:** Bipartite (User-Art)
- **Strict G2:** Hetero (+ Social)
- **Strict G3:** Hetero (+ Category)

Total Experiments: 6 models $\times$ 3 graphs $\times$ 2 protocols $=$ **36 Rigorous Runs**

Benchmark Results: Cold-Start / Full Protocol

Protocol: 80/20 Split (Random).
Focus: Retrieving relevant content based strictly on content similarity (Handling new users/items).

Model	Type	R@10	N@10	H@10	MRR
Content-Based / Baselines					
BGE-M3	**CB**	**0.0535**	**0.0315**	**0.0929**	**0.0416**
TF-IDF	CB	0.0508	0.0301	0.0951	0.0394
User-KNN (Pearson)	Memory CF	0.0412	0.0235	0.0762	0.0251
Graph G1 (Bipartite)					
LightGCL	Graph CL	0.0398	0.0249	0.0685	0.0313
XSimGCL	Graph CL	**0.0439**	0.0253	0.0694	0.0289
HGN	Hetero GNN	0.0262	0.0147	0.0451	0.0182
HCL	Hetero CL	0.0259	0.0164	0.0471	0.0214
SimGCL	Graph CL	0.0191	0.0115	0.0275	0.0135
Graph G2 (Hetero - User-User)					
LightGCL	Graph CL	**0.0436**	0.0261	0.0739	0.0313
XSimGCL	Graph CL	0.0432	0.0260	0.0688	0.0303
HGN	Hetero GNN	0.0369	0.0201	0.0596	0.0250
HCL	Hetero CL	0.0320	0.0192	0.0520	0.0233
SimGCL	Graph CL	0.0228	0.0131	0.0312	0.0144
Graph G3 (Category Hubs)					
LightGCL	Graph CL	**0.0413**	0.0256	0.0710	0.0311
XSimGCL	Graph CL	0.0379	0.0232	0.0643	0.0289
HGN	Hetero GNN	0.0299	0.0167	0.0464	0.0198
HCL	Hetero CL	0.0250	0.0152	0.0471	0.0205
SimGCL	Graph CL	0.0222	0.0122	0.0322	0.0137

Impact of Graph Structure

Question: Which graph topology yields the best performance?
Testing **HGN** on different graph types:

Graph	Structure Stats	R@10
G1 (Base)	13k Users, 30k Edges Density: 0.06%	0.0262
G2	**+61k Social Edges** Density: **0.064%**	**0.0369**
G3	+6 Categories 16k User-Cat Edges	0.0299

Insight: Denser is not always better. **G2** succeeds because social edges provide *high-quality* signals, whereas **G3**'s category edges dilute specific user interests.

Efficiency Analysis: Model Complexity and Training Time

Comparison: Training time per epoch (minutes:seconds) and parameter count.

Model	Params	Time (G1)	Time (G2)	Time (G3)
SimGCL	1,072,768	00:47	00:46	00:45
XSimGCL	1,072,770	**00:35**	**00:33**	**00:34**
LightGCL	1,072,768	00:46	00:45	00:45
MA-HCL	1,072,770	00:47	01:10	00:37
MA-HGN	1,273,986	01:29	02:23	00:35

Observations:

- **XSimGCL** is the most efficient method (fastest training time).
- **MA-HCL** maintains comparable efficiency to SimGCL/LightGCL on bipartite graphs (G1, G3) while handling heterogeneity.
- **MA-HGN** incurs significant computational cost on dense heterogeneous graphs (G2), taking **2-3x longer** than Contrastive Learning methods.

GNN Architecture

Comparing **GAT** (default), **GraphSAGE**, and **TransformerConv**.

Backbone	Full Protocol		Efficiency		
	R@10	N@10 i	Inf. (s)	Params	Train (s)
GAT	0.0262	0.0163	∼1.38	1.27M	20.23
SAGE	0.0303	0.0182	∼0.88	1.27M	19.77
TRANS	**0.0504**	**0.0266**	**1.71**	**1.47M**	19.76

Analysis:

- **Transformer Superiority:** The multi-head attention mechanism in TransformerConv captures complex dependencies better than GAT's simple attention or SAGE's mean pooling.
- **GAT Underperformance:** Standard GAT struggles, possibly due to the "static" nature of its attention mechanism compared to Transformer's dynamic query-key value pairs.
- **Consistent Ranking:** TransformerConv dominates in Cold Start scenarios.

Ablation Study: Impact of Meta-Paths

Question: How do different edge types contribute to performance?

Configuration	Recall@10	Drop (%)	Insight
Full Graph (G2)	**0.0369**	-	Best performance
w/o Social	0.0293	-20.6%	Social alignment is critical
w/o Category	0.0227	-38.5%	Categories provide broad signals

Conclusion:

- **Synergy:** Combining all edge types (Social + Category + Behavior) yields the highest Recall.
- **Category Impact:** Removing categories hurts performance the most, indicating they are essential for semantic alignment in cold-start scenarios.

Valid Semantic Enrichment: Category Hubs (G3)

Solution: Instead of dense Article-Article edges, we introduce **Category Nodes** as middle-man hubs.

The G3 Architecture

- **Compact:** Only 6 Category nodes.
- **Efficient:** Adds ∼12k edges.
- **Structure:** Dual-hop paths.
  - $User \leftrightarrow Category \leftrightarrow Article$

Role of Semantic Hubs

- **Connectivity:** Creates paths between users and articles even without direct history.
- **Signal Quality:** Captures thematic interest (e.g., "Sports") without the noise of random co-clicks.
- **Scalability:** Avoids the exploded edge count of item-item graphs.

Verdict: Category Hubs provide a scalable semantic alternative, effectively

Our Contributions: HGN & HCL

We propose two distinct architectures to handle the "VnExpress Heterogeneity":

1. HGN (Best for Structure)

- **Method:** Meta-path Aggregation + Attention Fusion.
- **Why:** Explicitly weighs "Social" vs "Interest".
- **Recall@10 (G2): 0.0369** (Highest Graph Model).

2. HCL (Contrastive)

- **Method:** Heterogeneity + Contrastive Loss.
- **Why:** Regularization against noise.
- **Recall@10 (G2):** 0.0320 (Stable).

Key Takeaway:

- **HGN** successfully exploits the Social graph (G2) to achieve the best structural performance (+40% over G1).
- **HCL** offers a lightweight alternative, though attention mechanisms

Evaluation Methodology & Validity

Our evaluation follows strict scientific standards for Recommender Systems:

- **Metric Choice:**
  - *Recall@10*: Measures system coverage.
  - *NDCG@10*: Rewards models that rank true items higher.
  - **MRR**: Standard for single-target news retrieval.
- **Dual Protocol Strategy:**
  - **Mainstream (LOO):** Temporal Leave-One-Out. Test on the *latest* interaction to prevent future leakage.
  - **Cold-Start (80/20):** Random user split. Train on 80% of users, test on 20% unseen users to measure zero-shot capability.

Ablation Study: Impact of GNN Depth (L)

Question: Does deeper graph propagation help? **Metric:** Recall@10 on Cold-Start (Full).

Model	L=1	L=2	L=3	L=4
LightGCL	0.032	**0.040**	**0.043**	**0.042**
XSimGCL	0.028	0.035	0.039	0.041
LightGCN	**0.031**	0.032	0.32	0.031
HGN	**0.033**	0.026	0.026	0.028
HCL	0.020	0.010	0.007	0.009
SimGCL	0.024	**0.027**	0.021	0.023

Insight:

- **LightGCN** suffers from *over-smoothing* (Performance drops after L=1).
- **XSimGCL** and **LightGCL** maintain robustness at deeper layers, capturing long-range dependencies effectively.

Performance Justification: Challenge of Sparsity

Is 3-4% Recall Low? No. Recommender performance heavily depends on **Graph Density**.

Dataset	#Users	#Items	Density	SOTA Recall@20
ML-10M	69k	10k	**0.982%**	$\sim$ 0.2613
Yelp	29k	24k	0.146%	$\sim$ 0.0793
Gowalla	50k	57k	0.040%	$\sim$ 0.1578
Amazon-Book	78k	77k	**0.037%**	**0.0585**
Tmall	47k	41k	0.119%	0.0528
Ours (G2)	**13k**	**3.6k**	**0.08%**	**0.0369** (@10)

Table 1: Comparison with Standard Benchmarks (LightGCN/SimGCL Results)

Our Dataset vs. Amazon-Book:

- Similar high sparsity ($\sim 0.0X\%$).
- News items have **shorter lifespans** than books/products, making "exact retrieval" (Recall) harder.

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Conclusion

Key Achievements:

- Built an end-to-end **Heterogeneous Graph Recommendation System** for Vietnamese News.
- Integrated **Social & Temporal** signals, resolving the user cold-start and sparsity problem.

Limitations:

- **Poor Performance:** MA-HGN MA-HCL yield bad results (Recall10 $\sim 3.7\%$).
- **Computational Cost:** Graph construction and neighborhood aggregation (GNN) are resource-intensive for large-scale graphs.
- **Offline Latency:** The current model is static; new articles require re-training or expensive incremental updates (no real-time adaptation).

Future Directions

Roadmap for Improvement:

- **Dynamic Graph Learning (dyHGCN):** Capture evolving user interests over time instead of static snapshots.
- **More Meta Path:** Use more meta paths to capture different types of relationships between entities.
- **Self-Supervised Learning (SSL):** Apply contrastive learning to de-noise the graph and improve robustness against sparsity.
- **Knowledge Distillation:** Compress the large GNN model into a lightweight student model for faster inference.

Thank You!

Q & A

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